

USER REQUIREMENTS SPECIFICATION
for
**TASKFORCE MONITORING SYSTEM
(TFMS)**

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1. INTRODUCTION

1.1 Purpose

The purpose of this User Requirements Specification (URS) document is to formally describe the functional, non-functional, and operational requirements for the development of the Task Force Monitoring System (TFMS) for the Faculty of Artificial Intelligence (FAI). The TFMS is intended to replace the current manual, spreadsheet-based approach to managing lecturer task force assignments, which poses challenges for accuracy, transparency, and workload balancing.

This document serves as a high-level contractual agreement between stakeholders—including system administrators, Heads of Department (HODs), Programme Coordinators, the Human Resource Officer (PSM), Faculty Management, and Academic staff—and the development team. It defines what the system shall achieve from the user's perspective, without prescribing technical design or implementation details.

The URS also provides a foundation for:

- subsequent preparation of the Software Requirements Specification (SRS);
- construction of design artefacts such as the Software Design Description (SDD);
- formulation of Software Test Descriptions and Test Results;
- project management planning, including scope, deliverables, and acceptance criteria.

1.2 Objectives

The TFMS is initiated to address the operational need for an integrated digital platform that streamlines the management of lecturer task force activities. The main objectives are:

- a) Establish the user requirements for the TFMS.
- b) Determine the scope of the system needed to meet the user requirements.
- c) Identify capacity and requirements for future expansion (e.g. additional faculties, campuses).
- d) Provide the basis for the development, testing and acceptance of the TFMS.

1.3 Scope

1.3.1 System Scope

The TFMS will provide comprehensive support for the management of academic task forces, including assignment, monitoring, approval workflows, and workload fairness assessment. The system includes:

- a) A centralised database of staff, task forces, and membership information.
- b) Secure login and role-based access control.
- c) Interfaces for System Administrator, HOD/Task Force Owner, PSM/HR Officer, Lecturers, and Faculty Management.
- d) Automated weightage calculation and fairness indicators.
- e) Reporting and analytics features for operational and strategic planning.
- f) The system is limited to the administration of task force activities only. It does not handle teaching workload, research KPI tracking, or performance appraisal—though it may provide summaries useful for those processes.

1.3.2 Interfaces with Users

The TFMS will interface with:

- **System Administrator** – manages master data (staff and task forces), system parameters, and user accounts.
- **HOD / Task Force Owner** – proposes and updates task force membership, handles departmental allocations, and reviews staff workload.
- **PSM / HR Officer** – oversees faculty-level balancing, approval workflows, and reporting.
- **Lecturers** – view their task force portfolio, history, and workload status.
- **Faculty Management** – accesses high-level dashboards and strategic summaries.

The system may later interface with external services such as university single sign-on (SSO), HR systems, or academic records if required.

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1.3.3 Software Documentation

To support system operation and long-term maintenance, the following documentation will be produced:

- a) SDP – Software Development Plan (Update the existing document)
- b) SRS — Software Requirements Specification
- c) SDD — Software Design Descriptions
- d) STD — Software Test Descriptions
- e) STR — Software Test Results
- f) SDF – Software Development Files

1.4 Definitions, Acronyms and Abbreviations

1.4.1 Definitions

- **Shall** – Indicates a mandatory requirement that must be implemented.
- **Should** – Indicates a desirable requirement that has significant merit but may not be mandatory.
- **May** – Indicates an optional feature or function.
- **Task Force** – A committee, working group, or special assignment involving academic staff for academic, administrative, research, quality assurance, or strategic duties.
- **Weightage** – A numerical metric that reflects the workload contribution of a task force to a staff member.
- **Under-loaded / Balanced / Over-loaded** – Classification based on whether a lecturer's total task force weightage falls below, within, or above configured thresholds.
- **Active Staff** – Staff currently serving in the faculty and eligible for task force assignment.

1.5 Overview of the Document

This document contains:

- **Section 1:** Introduction to the TFMS, purpose, scope, objectives, and definitions.
- **Section 2:** Overview of user requirements, system context, and constraints.
- **Section 3:** Detailed functional requirements categorised by user roles and system components.
- **Section 4:** Quality and non-functional requirements (security, reliability, performance).
- **Section 5–7:** Environment, design constraints, and system capacity expectations.

2. OVERVIEW OF REQUIREMENTS

2.1 Background

The Faculty of Artificial Intelligence (FAI) operates numerous task forces each academic year, covering critical areas such as curriculum development, accreditation, research enhancement, quality assurance (MQA compliance), examination management, postgraduate coordination, academic integrity, industrial collaboration, and strategic planning.

Traditionally, task force information has been maintained using spreadsheets or manual documents distributed among departments. Although functional at a small scale, this method introduces several systemic challenges.

2.1.1 Operational Problems in the Current Manual Process

- a) **Lack of Real-Time Visibility**
Updates to task force membership are often delayed because multiple spreadsheet versions circulate between departments, leading to data inconsistencies.
- b) **Unfair Distribution of Workload**
Without automated weightage calculation, some lecturers become overloaded while others remain under-utilized.
- c) **Inefficient Approval Workflow**
HODs propose members manually, requiring multiple rounds of email communication before final approval by PSM or faculty management.
- d) **Poor Data Traceability**
Justifications for changes (e.g., removing or adding certain members) are not systematically documented.
- e) **Fragmented Historical Records**
Lecturers often need documentation of task force involvement for appraisal, promotion, or performance review, but historical data is difficult to retrieve.
- f) **No Analytical Capability**
Manual spreadsheets cannot automatically generate dashboards, charts, or workload analyses for informed decision-making.

2.1.2 Rationale for TFMS Development

The Task Force Monitoring System (TFMS) is proposed to remedy the above shortcomings by providing a centralised, secure, and role-based digital environment. It shall support transparent workload balancing, streamlined membership updates, automated weightage computation, and reliable reporting that aligns with good governance practices.

2.2 Stakeholders and System Interfaces

The TFMS will serve multiple user categories. Each group has specific needs that directly influence the system requirements.

2.2.1 System Administrator

 Responsible for configuring the system, maintaining master data, managing user accounts, and supporting system operations. Interfaces required:

- Staff data management interface
- Task force configuration interface
- System settings and weightage rules
- Audit log viewer

2.2.2 Head of Department (HOD) / Task Force Owner

 Manages proposed membership lists, updates roles, and ensures fair workload distribution within their department. Key interfaces:

- Departmental member assignment dashboard
- Lecturer workload analytics
- Membership proposal submission page

2.2.3 PSM / Human Resource Officer

Oversees faculty-wide task force balancing and approves final membership lists. Interfaces required:

- Faculty-wide workload monitoring panel
- Approval workflow screen
- Reporting and export functions

2.2.4 Lecturers

Lecturers can access personal dashboards that show task force participation, weightage, and history. Interfaces include:

- Personal portfolio view
 - Historical task force records
 - Printable summaries
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2.2.5 Faculty Management (Dean / Deputy Dean)

Requires strategic insights into task force distribution patterns, fairness metrics, and departmental load comparison. Interfaces required:

- Executive dashboards
- High-level trends and analytics
- Drill-down reporting

2.3 System Functions

This section summarises the major system functions defined from a user standpoint. Each function is further elaborated in Chapter 3 (Functional Requirements).

2.3.1 Authentication and Access Control

- Secure login for all user roles
- Password policy enforcement
- Role-based access control to segregate features
- Session-based authentication management

2.3.2 Staff Master Data Management

- Add/edit staff details (name, ID, department, grade, status)
- Define academic roles (HOD, Programme Coordinator, Task Force Chair)
- Activate/deactivate staff
- Bulk import functionality (optional future enhancement)

2.3.3 Task Force Master Data Management

- Create task forces with metadata (name, category, default weightage, owner, active dates)
- Edit or deactivate task forces
- Tag task forces by department or faculty focus area
- Assign default weightage or role-specific weightage

2.3.4 Task Force Membership Management

- Add/remove/update task force members
- Assign member roles (Chair, Member, Secretariat)
- Validate eligibility (e.g., active staff, department matching)
- Provide automated suggestions for under-loaded lecturers
- Record justification for changes

2.3.5 Workload Weightage Calculation and Fairness Engine

- Sum total weightage per lecturer
- Classify staff as *under-loaded*, *balanced*, or *over-loaded*
- Highlight members exceeding configured thresholds
- Allow HOD or PSM to override fairness rules with justification



2.3.6 Approval Workflow

- HOD submits proposed lists to PSM
- PSM reviews, adjusts, and approves
- System locks records after final approval
- Exceptional changes recorded with timestamp

2.3.7 Reporting and Analytics

- Department-level task force lists
- Lecturer workload summary tables
- Faculty-wide fairness distribution
- Visualization: pie charts, bar graphs, ranking tables
- Export to Excel/PDF

2.3.8 Notifications and Alerts

- Deadline reminders set by PSM
- Alerts on over-loaded staff
- Notifications of pending approvals
- Change logs and membership updates



2.4 User Characteristics

2.4.1 General User Population

The TFMS will be used by approximately 80–150 academic and administrative staff. Users are expected to have:

- basic ICT literacy;
- familiarity with web applications;
- experience handling faculty-level academic data.

2.4.2 Special Considerations

- User training will be minimal due to intuitive interface design.
- Accessibility features (e.g., clear UI, colour contrast) shall be considered.
- The system will primarily be used during office hours, but should support off-site access.

2.5 General Constraints

2.5.1 Institutional Constraints

- System must align with university policies on workload distribution.
- Data storage must comply with institutional data governance regulations.
- Approval workflows must follow faculty administrative procedures.

2.5.2 Technical Constraints

- Web-based system accessible via modern browsers.
- Backend must support relational database management.
- All configurations (weightage thresholds, roles, categories) must be changeable without code modification.

2.5.3 Security Constraints

- All sensitive data must be encrypted at rest or in transit.
- Passwords must be hashed.
- Audit logging is mandatory.
- Role-based access control is required.

2.5.4 Operational Constraints

- High system availability during faculty administrative cycles (e.g., beginning of semesters).
- Minimal downtime during task force submission periods.

2.6 System Boundaries

Inside System Boundary

- Task force creation
- Membership management
- Weightage calculation
- Fairness analytics
- Reports and dashboards
- Approval workflows
- Historical data storage

Outside System Boundary

- Staff performance evaluation system (HRMS)
- Teaching load calculation
- Research KPI tracking system
- Payroll and attendance system
- E-learning and LMS integration

(Integration may be added in future expansions.)

3. FUNCTIONAL REQUIREMENTS

Chapter 3 specifies the complete set of functional requirements of the Task Force Monitoring System (TFMS). Requirements are organised according to user roles, system components, and logical processing flows. Each requirement is written using explicit, verifiable, and testable language following the “*The system shall...*” format.

3.1 Overall Description of Functional Requirements

The TFMS provides role-based functionalities for multiple stakeholders, including System Administrators, Heads of Departments (HODs), Task Force Owners, PSM/HR Officers, Lecturers, and Faculty Management. The system supports authentication, master data management, membership assignment, workload calculation, fairness analytics, approval workflows, reporting, notifications, and audit logging.

The functional requirements are presented under the following categories:

- a) **Login and Security Requirements**
- b) **System Administrator Requirements**
- c) **HOD / Task Force Owner Requirements**
- d) **PSM / HR Officer Requirements**
- e) **Lecturer Requirements**
- f) **Faculty Management Requirements**
- g) **System-wide Processing Requirements**
- h) **Audit Trail and Logging Requirements**
- i) **Search, Filtering, Sorting, and Export Requirements**

3.2 Functional Requirement Categories

3.2.1 Login and Security Requirements

3.2.1.1 Authentication Requirements

The system shall:

- a) Provide secure login for all users.
- b) Require username and password according to the following policy:
 - i. minimum 8 characters,
 - ii. maximum 16 characters,
 - iii. must include alphanumeric characters,
 - iv. may include special characters.
- c) Lock an account after three (3) failed login attempts.
- d) Allow administrators to unlock accounts.
- e) Provide “Forgot Password” functionality using email verification or identity check.
- f) Require first-time users to change their initial password.
- g) Track login and logout events in the audit log.

3.2.1.2 Role-Based Access Control

The system shall:

- a) Restrict access to functions based on user role (Admin, HOD, PSM, Lecturer, Management).
- b) Prevent users from accessing data outside their authorised scope.
- c) Display role-appropriate dashboards and menus.

3.2.2 System Administrator Functional Requirements

3.2.2.1 Staff Master Data Management

The system shall:

- a) Allow Admin to add, edit, or deactivate staff records. 
- b) Store staff details:
 - i. Staff ID
 - ii. Name & title
 - iii. Department 
 - iv. Grade (Professor/Assoc. Prof/Senior Lecturer/Lecturer) 
 - v. Employment status (permanent, contract, visiting)
 - vi. Role flags (e.g., HOD, Task Force Chair, Programme Coordinator) 
- c) Prevent duplication of staff IDs. 
- d) Allow bulk import of staff via spreadsheet (future enhancement).

3.2.2.2 Task Force Master Data Management

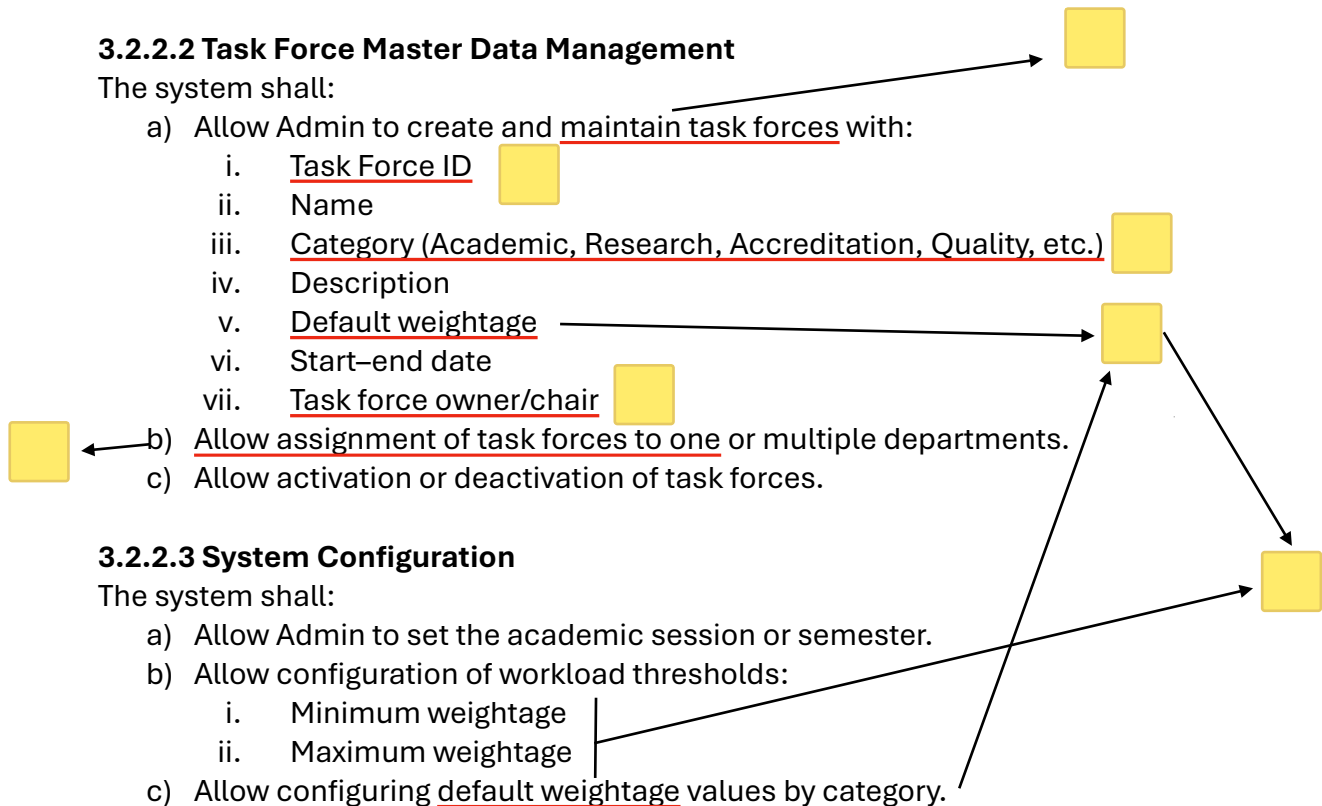
The system shall:

- a) Allow Admin to create and maintain task forces with:
 - i. Task Force ID 
 - ii. Name
 - iii. Category (Academic, Research, Accreditation, Quality, etc.) 
 - iv. Description
 - v. Default weightage 
 - vi. Start-end date
 - vii. Task force owner/chair 
- b) Allow assignment of task forces to one or multiple departments. 
- c) Allow activation or deactivation of task forces.

3.2.2.3 System Configuration

The system shall:

- a) Allow Admin to set the academic session or semester.
- b) Allow configuration of workload thresholds:
 - i. Minimum weightage
 - ii. Maximum weightage
- c) Allow configuring default weightage values by category.



3.2.2.4 User Account Management

The system shall allow Admin to:

- a) Create user accounts.
- b) Assign user roles.
- c) Reset passwords.
- d) Unlock accounts.

3.2.2.5 Audit Log Access

The system shall:

- a) Provide Admin with access to audit logs filtered by:
 - i. date range,
 - ii. user,
 - iii. action type.
- b) Store create/update/delete events.

3.2.3 HOD / Task Force Owner Requirements

3.2.3.1 Task Force Viewing and Filtering

The system shall allow HOD to:

- a) View all task forces under their department/ownership.
- b) View current members and roles of each task force.
- c) Filter by category, session, department.

3.2.3.2 Membership Management

The system shall allow HOD to:

- a) Add members to task forces.
- b) Remove members from task forces.
- c) Assign roles (Chair, Member, Secretariat).
- d) View individual lecturer workload and status.
- e) View system-generated suggestions based on under-loaded lecturers.

3.2.3.3 Workload and Fairness Checking

The system shall:

- a) Compute total weightage for each lecturer.
- b) Highlight lecturers as:
 - i. Under-loaded
 - ii. Balanced
 - iii. Overloaded
- c) Allow HOD to override recommended assignments with justification.

3.2.3.4 Submission for Department-Level Approval

The system shall:

- a) Allow HOD to submit the proposed task force list to PSM.
- b) Notify HOD about approval or rejection.
- c) Prevent further changes once the list is locked by PSM.

3.2.4 PSM / HR Officer Requirements

3.2.4.1 Faculty-Wide Task Force Overview

The system shall allow PSM to:

- a) View all task forces across all departments.
- b) View membership, categories, active dates, and owner.

3.2.4.2 Workload Balancing and Adjustment

The system shall:

- a) Display faculty-wide workload distributions.
- b) Allow PSM to adjust assignments across departments.
- c) Recalculate workload instantly after changes.

3.2.4.3 Approval Workflow

The system shall allow PSM to:

- a) Approve or reject departmental submissions.
- b) Lock the task force list for a semester.
- c) Approve exceptional changes after lock, with justification.

3.2.4.4 Faculty Reporting

The system shall allow PSM to generate:

- a) Task force membership lists.
- b) Lecturer workload tables.
- c) Under/overload summaries.
- d) Export to Excel/PDF.

3.2.5 Lecturer Requirements

The system shall allow Lecturers to:

- a) Log in and access a personal dashboard.
- b) View assigned task forces with:
 - i. Role
 - ii. Category
 - iii. Weightage
- c) View total weightage and load status.
- d) View historical records for past semesters.
- e) Download/print a summary for appraisal purposes.
- f) Submit remarks (e.g., sabbatical, heavy research) to HOD/PSM for consideration.

3.2.6 Faculty Management Requirements

The system shall allow management to:

- a) View executive dashboards summarising:
 - I. Task Force Distribution

- II. workload fairness → 
- III. Department comparison → 
- b) Drill down to department or individual level.
- c) Export summary reports.

3.2.7 System-Wide Processing Requirements

The system shall:

- a) Perform weightage calculation for each lecturer.
- b) Detect workload imbalance.
- c) Support real-time updates across departments.
- d) Validate members based on employment status.
- e) Prevent assigning inactive or non-eligible staff.
- f) Support automated member suggestions.

3.2.8 Reporting, Search, and Export Requirements

The system shall:

- a) Provide advanced search (by staff, task force, category, session, department).
- b) Support sorting and filtering for all tables.
- c) Generate reports in PDF and Excel format. 
- d) Display graphs (bar charts, pie charts, distribution graphs).
- e) Allow generating appraisal-ready summaries for lecturers.

3.2.9 Audit Trail Requirements

The system shall:

- a) Log all create/update/delete actions.
- b) Log approvals, rejections, lock/unlock events.
- c) Store user identity, timestamp, and justification.
- d) Allow admin to export audit logs for compliance.

4. QUALITY REQUIREMENTS

4.1 Reliability

- a) The TFMS shall be available at least 99% of the time during office hours, excluding scheduled maintenance.
- b) The system shall ensure that no data is lost due to normal user operations; all confirmed updates shall be stored reliably in the database.

4.2 Maintainability

- a) The system shall be designed to allow changes in weightage rules and thresholds without requiring code modification (i.e. via configuration tables).
- b) Maintenance downtime shall not usually exceed six hours per maintenance window.

4.3 Performance

- a) The system shall support concurrent access by multiple lecturers, HODs and administrative staff.
- b) For typical queries (e.g. listing task forces for a department), the system response time should not exceed 3 seconds under normal load.

4.4 Security

- a) Access to system functions shall be restricted based on user roles (Administrator, HOD, PSM, Lecturer, Management).
- b) All passwords shall be stored using secure hashing mechanisms.
- c) The system shall log key user actions such as login, create/update/delete of task forces and memberships, and approval or locking actions.

4.5 Expandability

- a) The system should be extendable to include additional faculties within the university in the future.
- b) The system should be designed to integrate with other university systems (e.g. staff directory, single sign-on) where available.

5. ENVIRONMENT

5.1 Hardware

The system shall be accessible via desktop or notebook computers running Windows 10, Windows 11 or equivalent operating systems, and via modern tablets where feasible.

5.2 Software

- a) The system shall operate as a web application accessible using standard web browsers (e.g. Chrome, Edge, Firefox) without requiring client-side installation.
- b) The server-side software may be developed using any programming language and framework agreed by the development team and supervisor, provided it supports secure web operations and database connectivity.

6. DESIGN CONSTRAINTS

- a) Object-oriented development (OOD) techniques with UML notation shall be applied to this project.
- b) A relational database management system (RDBMS) shall be used to store all persistent data.
- c) The user interface shall follow the university's basic branding guidelines where applicable.

7. CAPACITY

- a) Initially, the system shall cater to all programmes belonging to FAI and up to a few hundred academic staff and task forces per academic year.
- b) The system design should allow future scaling to support additional faculties, campuses or institutes without major redesign.

Example of Taskforce Monitoring in Spreadsheet

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4						
5	Bil	Nama Pensyarah	JK/TF 1	JK/TF 2	JK/TF 3	JK/TF 4
5						
26	Dr. Noor Hafizah binti Hassan	Task Force Smart Classroom (A)	Task Force Curriculum Review Program Msc IA	Technical Program Chair (TPC)JICIBEM 2020'	Task Force Datathon 2020	Task Force IPDOCS 2021
30						
27	Dr. Mohd Syahid bin Mohd Anuar	Task Force Smart Classroom (A)				
31						
28	Dr Suriyani bt Mohd Sam	Task Force AKA 2019 (A)	Task Force Curriculum Review Program MSc. CSE (A)	Technical Program Chair (TPC) International Conf. on Telematics and Future Generation Networks (TAFGEN2020)	Technical Program Chair (TPC) International Conference of Advanced Research on Renewable Energy For Universal Sustainability (ARUS2021)	
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